

Lab 1:

Objective: The goal of this lab is to reinforce the fundamental concepts of dynamic memory allocation, object-oriented programming, and deep copying in C++. Students will gain hands-on experience with constructors, destructors, copy constructors, and array-based dynamic memory management while developing a custom integer list class.

Problem Statement: In this lab, you are required to implement a dynamic integer list class in C++ that manages an array of integers.

Steps to follow:

1. Write a search operation that searches for a given value in the list. If the value exists, return the index of its first occurrence. If the value is not found, return -1 and display an appropriate message.
2. Write a driver program to test the search function and show that your search operation works correctly.
 1. Add 6 numbers to the list.
 2. Search for different values, including one that exists and one that doesn't exist.

Google Classroom submission:

1. Your source code (1 .sln or .cpp file)
2. A screenshot that shows your test (Do not zip)